Sharp worst-case factors for randomly pivoted Cholesky and LU

Matthew J. Colbrook*

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Abstract

For every positive integer r, an explicit one-parameter family of real, entrywise-positive, positive-definite matrices of order $r+1$ makes the expected trace error of r randomly pivoted Cholesky steps approach 2^r times the optimal rank-r trace error. The same family makes the expected squared Frobenius error of r randomly pivoted LU steps approach 4^r times the optimal rank-r squared Frobenius error. Thus both previously established universal factors are sharp as suprema. The same suprema hold on positive-definite correlation matrices when the dimension is unrestricted. In particular, a polynomial-in-r factor for same-rank RPCholesky and a 2^r factor for same-rank RPLU are impossible without additional hypotheses. The proof combines scale-separated Cauchy–Binet expansions with an exact cancellation over pivot histories. A separate 2×2 rational example already disproves the literal 2^r RPLU bound at $r = 1$. Exact rational verification certificates accompany the proof.

1 Algorithms and results

For a positive-semidefinite matrix A, write

$$\tau_r(A) = \sum_{j>r} \lambda_j(A), \qquad \lambda_1(A) \geq \cdots \geq \lambda_n(A) \geq 0.$$

Randomly pivoted Cholesky starts from $R_0 = A$. Given a nonzero residual R_s, it chooses j with probability $(R_s)_{jj}/\operatorname{tr} R_s$ and sets

$$R_{s+1} = R_s - \frac{(R_s)_{:j}(R_s)_{j:}}{(R_s)_{jj}}. \tag{1}$$

The residual is positive semidefinite, so its trace is also its nuclear norm. The known universal bound is

$$\mathbb{E} \operatorname{tr} R_r \leq 2^r \tau_r(A) \qquad [1, \text{Lemma 5.5}]. \tag{2}$$

*Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Cambridge, United Kingdom. Email: m.colbrook@damtp.cam.ac.uk.

Randomly pivoted LU starts from $B_0 = A$, now allowing a general rectangular complex matrix. It chooses (i, j) with probability $|(B_s)_{ij}|^2 / \|B_s\|_{\mathrm{F}}^2$ and sets

$$B_{s+1} = B_s - \frac{(B_s)_{:j}(B_s)_{i:}}{(B_s)_{ij}}. \tag{3}$$

A zero entry has zero selection probability. If the residual becomes zero, the algorithm can stop. If A_r denotes a best rank-r approximation, the known bound is

$$\mathbb{E}\,\|B_r\|_{\mathrm{F}}^2 \le 4^r \,\|A - A_r\|_{\mathrm{F}}^2 \qquad \text{[2, Theorem 3]}. \tag{4}$$

Gilles and Wilber conjecture that the factor 4^r can be replaced by 2^r [2, discussion after Theorem 3]. The two questions addressed here are same-rank approximation questions: exactly r pivots are compared with the best rank-r error. They correspond to the polynomial-factor RPCholesky question and the 2^r RPLU question catalogued as RA-02 and RA-03, respectively, in [3]. All algorithmic statements below are in exact arithmetic.

For $r \ge 1$, put $n = r + 1$. For $0 < t < 1$, define

$$\begin{gathered} L(t)_{ij} = \begin{cases} 1, & i = j, \\ t^j/(i+j), & i > j, \\ 0, & i < j, \end{cases} \qquad 1 \le i, j \le n, \\ \epsilon = t^{n^2}, \qquad D = \operatorname{diag}(1, \epsilon, \ldots, \epsilon^{n-1}), \qquad A_r(t) = L(t) D L(t)^{\mathsf{T}}. \end{gathered} \tag{5}$$

The matrix is positive definite for every $t > 0$. Its entries are positive: for each off-diagonal entry the contribution of the first column of L is already positive. Rational t gives a rational matrix.

Theorem 1 (Sharp lower-bound family)**.** *Fix $r \ge 1$. Apply r steps of the two algorithms to $A = A_r(t)$ in* (5)*. Then*

$$\lim_{t \downarrow 0} \frac{\mathbb{E}\operatorname{tr} R_r}{\tau_r(A)} = 2^r, \qquad \lim_{t \downarrow 0} \frac{\mathbb{E}\,\|B_r\|_{\mathrm{F}}^2}{\|A - A_r^{\mathrm{best}}\|_{\mathrm{F}}^2} = 4^r, \tag{6}$$

where A_r^{best} denotes a best rank-r approximation of A. Every square submatrix of $A_r(t)$ is nonsingular for all sufficiently small $t > 0$, with the threshold allowed to depend on r. Thus every ordered pivot history appearing in the proof has positive probability.

Corollary 2 (Worst-case constants)**.** *For each $r \ge 1$, the supremum of the RPCholesky ratio over all positive-semidefinite inputs with nonzero rank-r tail is 2^r. The supremum of the RPLU squared-error ratio over all matrix inputs with nonzero rank-r error is 4^r. Restricting the lower-bound examples to real positive-definite, entrywise-positive matrices does not change either supremum.*

Proof. The upper bounds are (2) and (4); Theorem 1 gives the matching lower bounds. □

Corollary 3 (Consequences for the proposed improvements)**.** *There are no fixed constants $C > 0$ and $p \ge 0$ such that $\mathbb{E}\operatorname{tr} R_r \le C r^p \tau_r(A)$ for every r and every positive-semidefinite A. For RPLU, no bound of the form $\mathbb{E}\,\|B_r\|_{\mathrm{F}}^2 \le C 2^r \,\|A - A_r\|_{\mathrm{F}}^2$ holds with C independent of r. More generally, no factor $q(r)2^r$, with q a fixed nonnegative polynomial, can replace 4^r uniformly.*

Proof. For the first statement, choose r so that $2^{r-1} > C r^p$, and then use Theorem 1 with t sufficiently small. For the second and third statements, a zero proposed factor is immediately impossible. Otherwise, $4^r/(2^r q(r)) = 2^r/q(r)$ is unbounded; again choose r first and then t. No limit uniform in r is needed. □

2 A two-dimensional RPLU counterexample

The literal factor-2^r inequality fails even at $r = 1$. Take

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \tag{7}$$

Its singular values are 3 and 1, so the optimal rank-one squared Frobenius error is 1. The total squared entry norm is 10. Each diagonal pivot has probability 2/5 and leaves one nonzero residual entry of magnitude 3/2. Each off-diagonal pivot has probability 1/10 and leaves one nonzero residual entry of magnitude 3. Therefore

$$\mathbb{E}\,\|B_1\|_{\mathrm{F}}^2 = 2\left(\frac{2}{5}\frac{9}{4}\right) + 2\left(\frac{1}{10}9\right) = \frac{18}{5} > 2\,\|A - A_1\|_{\mathrm{F}}^2\,. \tag{8}$$

This calculation requires no limiting argument and no numerical eigensolver.

More generally, if every entry of a nonsingular 2×2 matrix is nonzero, a pivot at (i, j) leaves the complementary entry with magnitude $|\det A|/|A_{ij}|$. Summing over the four pivot probabilities gives

$$\mathbb{E}\,\|B_1\|_{\mathrm{F}}^2 = \frac{4|\det A|^2}{\|A\|_{\mathrm{F}}^2}, \qquad \frac{\mathbb{E}\,\|B_1\|_{\mathrm{F}}^2}{\sigma_2(A)^2} = \frac{4\sigma_1(A)^2}{\sigma_1(A)^2 + \sigma_2(A)^2}.$$

The latter ratio tends to 4 when the singular-value ratio tends to infinity. The rest of the note establishes sharpness at every rank, not just at rank one.

3 Prefix minors and scale separation

Throughout the proof, r (and hence n) is fixed and $t \downarrow 0$. All implicit constants may depend on the fixed dimension. A submatrix indexed by sets is ordered increasingly unless an appended row or column order is specified. For $S \subseteq \{1, \dots, n\}$ with $|S| = s$, set

$$M_s(S) = L[S, 1\!:\!s], \qquad d_s(S) = \sum_{j \in \{1,\dots,s\} \setminus S} j.$$

An empty determinant is 1.

Lemma 4 (Prefix-row minors)**.** *For every such S,*

$$\det M_s(S) = \kappa_s(S) t^{d_s(S)}(1 + O(t)), \qquad \kappa_s(S) \neq 0. \tag{9}$$

For $s \geq 1$, $M_s(S)$ is invertible for sufficiently small t, and $\|M_s(S)^{-1}\|_2 = O(t^{-s})$.

Proof. Let $P = S \cap \{1, \dots, s\}$, $I = S \setminus \{1, \dots, s\}$, and $J = \{1, \dots, s\} \setminus S$. Thus $|I| = |J|$. Group the rows as (P, I) and the columns as (P, J). Multiply each column indexed by $j \in J$ by t^{-j}, leaving columns in P unchanged. The resulting matrix has a finite limit of the form

$$\begin{pmatrix} I & F \\ 0 & C \end{pmatrix}, \qquad C = \left[\frac{1}{i + j}\right]_{i \in I,\, j \in J},$$

where F is finite. Indeed, $L[P, P] \to I$, $L[I, P] \to 0$, and the scaled $L[I, J]$ equals C exactly. In the block $L[P, J]$, no diagonal entry occurs because P and J are disjoint, so column scaling leaves bounded entries.

The Cauchy matrix C is nonsingular. For distinct increasing x_a and y_b with positive sums, the elementary Cauchy determinant identity is

$$\det\left[\frac{1}{x_a+y_b}\right]_{a,b=1}^{m} = \frac{\prod_{a<a'}(x_{a'}-x_a)\prod_{b<b'}(y_{b'}-y_b)}{\prod_{a,b}(x_a+y_b)} \neq 0.$$

The empty block has determinant 1. Undoing the column scaling gives (9), including a possible sign from column ordering. The scaled matrix and its inverse remain bounded. Undoing the scaling in the inverse costs at most t^{-s}, proving the norm bound. ☐

Lemma 5 (Dominance of the leading spectral scales). *For row and column sets I, J of cardinality $s \leq n-1$,*

$$\det A[I,J] \sim \epsilon^{s(s-1)/2} \det M_s(I) \det M_s(J). \tag{10}$$

For $s = n$, the corresponding equality is exact, with $M_n = L$ and $\det L = 1$. Consequently, every square minor of A is nonzero for all sufficiently small positive t.

Proof. Cauchy–Binet gives

$$\det A[I,J] = \sum_{\substack{T\subseteq\{1,\ldots,n\}\\ |T|=s}} \det L[I,T] \det L[J,T]\, \epsilon^{\sum_{j\in T}(j-1)}.$$

The smallest power of ϵ occurs uniquely at $T = \{1,\ldots,s\}$ and is $s(s-1)/2$. All other powers are at least one larger. Since $L(t)$ remains bounded, all minors in the finite sum remain bounded. Thus

$$\det A[I,J] = \epsilon^{s(s-1)/2} \left(\det M_s(I) \det M_s(J) + O(\epsilon)\right).$$

Lemma 4 shows that the leading product has nonzero coefficient and t-degree at most $s(s+1)$. Since $\epsilon = t^{n^2}$ and $n^2 > s(s+1)$ for $s \leq n-1$, the remainder divided by that product tends to zero. If $s = n$, the sum has only one term. There are finitely many minors in a fixed dimension, so their eventual nonvanishing holds simultaneously. ☐

4 Residual normalizers

For $|S| = s < n$, let $l_{s+1} = L[:, s+1]$ and define

$$c_s(S) = M_s(S)^{-1} l_{s+1}[S], \qquad w_s(S) = l_{s+1} - L[:, 1\!:\!s]c_s(S), \qquad h_s(S) = \|w_s(S)\|_2^2. \tag{11}$$

For $s = 0$, the coefficient vector is empty and $w_0(\varnothing) = l_1$. The vector $w_s(S)$ vanishes on S.

Lemma 6 (A binary limiting rule). *For each fixed S of size $s < n$,*

$$\lim_{t\downarrow 0} \frac{1}{h_s(S)} = \mathbf{1}_{\{s+1\notin S\}}. \tag{12}$$

Moreover, $h_s(S) \geq 1/4$ for all sufficiently small t.

Proof. Because $L \to I$, its smallest singular value is at least $1/2$ eventually. The vector $w_s(S)$ equals L times a vector whose first s entries are $-c_s(S)$, whose $(s+1)$st entry is 1, and whose remaining entries vanish. It follows that

$$h_s(S) \geq \frac{1}{4}\left(1 + \|c_s(S)\|_2^2\right). \tag{13}$$

If $s+1 \notin S$, every entry of $l_{s+1}[S]$ is either zero or $O(t^{s+1})$. Lemma 4 therefore gives $c_s(S) = O(t)$, so $w_s(S) \to e_{s+1}$ and $h_s(S) \to 1$.

If $s+1 \in S$, the row of $M_s(S)$ indexed by $s+1$ has norm $O(t)$, whereas the same entry of $l_{s+1}[S]$ is exactly 1. The equation defining $c_s(S)$ forces $\|c_s(S)\|_2 \geq c/t$ for a positive constant c and small t. Equation (13) then gives $h_s(S) \to \infty$. The empty-set case is included in the first alternative. $\square$

For equally sized row and column sets I, J, define the full, zero-padded interpolatory residual

$$\mathcal{R}(I,J) = A - A[:,J]A[I,J]^{-1}A[I,:].$$

For empty sets this means $\mathcal{R}(\varnothing,\varnothing) = A$. Its rows in I and columns in J vanish. After any nonzero-pivot history with selected row set I and column set J, Gaussian elimination gives this residual, independently of the order within the selected sets. In the Cholesky case $I = J$.

Lemma 7 (Asymptotic normalizers)**.** *For $|I| = |J| = s < n$, one has*

$$T_s(S) := \operatorname{tr}\mathcal{R}(S,S) \sim \epsilon^s h_s(S), \tag{14}$$
$$Q_s(I,J) := \|\mathcal{R}(I,J)\|_{\mathrm{F}}^2 \sim \epsilon^{2s} h_s(I) h_s(J). \tag{15}$$

In particular,

$$\frac{\epsilon^s}{T_s(S)} \longrightarrow \mathbf{1}_{\{s+1 \notin S\}}, \tag{16}$$
$$\frac{\epsilon^{2s}}{Q_s(I,J)} \longrightarrow \mathbf{1}_{\{s+1 \notin I\}} \mathbf{1}_{\{s+1 \notin J\}}. \tag{17}$$

Proof. For $i \notin I$ and $j \notin J$, the bordered determinant identity, with i and j appended to the existing orders, gives

$$\mathcal{R}(I,J)_{ij} = \frac{\det A[(I,i),(J,j)]}{\det A[I,J]}.$$

Apply Lemma 5 to numerator and denominator. The ratio of the powers of ϵ is ϵ^s. A second bordered determinant identity gives

$$\det L[(I,i), 1\!:\!s+1] = \det M_s(I)\, w_s(I)_i,$$

and the analogous column identity gives $\det M_s(J)\, w_s(J)_j$. Consequently,

$$\mathcal{R}(I,J)_{ij} = \epsilon^s w_s(I)_i w_s(J)_j (1 + o(1)). \tag{18}$$

The two components in this formula are nonzero for sufficiently small t, by Lemma 4 applied to the bordered prefix minors. For a fixed dimension, the relative $o(1)$ is uniform over the finitely many relevant entries and sets.

For principal residuals, the diagonal leading terms in (18) are $\epsilon^s w_s(S)_i^2$. Summing them proves (14). Squaring and summing all entries proves (15). These are relative asymptotics even when a vector w_s diverges: the maximum relative error tends to zero before the weighted finite sum is taken. Finally use Lemma 6; its lower bound on h_s justifies inversion also in the diverging cases. $\square$

5 Exact cancellation over pivot histories

The next identities hold before taking any limit. Recall that $n = r + 1$. For a Cholesky history $(j_1, \ldots, j_r)$ of distinct indices, let $S_s = \{j_1, \ldots, j_s\}$. For an LU history, define similarly $I_s = \{i_1, \ldots, i_s\}$ and $J_s = \{j_1, \ldots, j_s\}$.

Lemma 8 (History identities)**.** *For an $(r+1) \times (r+1)$ positive-definite input with all square minors nonzero, the expectations after r pivots satisfy*

$$\mathbb{E} \operatorname{tr} R_r = \det A \sum_{(j_1,\ldots,j_r)} \prod_{s=0}^{r-1} \frac{1}{T_s(S_s)}, \tag{19}$$

$$\mathbb{E} \|B_r\|_{\mathrm{F}}^2 = |\det A|^2 \sum_{\substack{(i_1,\ldots,i_r)\\(j_1,\ldots,j_r)}} \prod_{s=0}^{r-1} \frac{1}{Q_s(I_s, J_s)}. \tag{20}$$

The sums range over ordered lists of distinct indices; in the second sum row indices and column indices are separately distinct.

Proof. In Cholesky, the product of the r pivot values is the principal determinant $\det A[S_r, S_r]$. The probability of the history is this product divided by $\prod_{s=0}^{r-1} T_s(S_s)$. Only one diagonal residual entry remains after r pivots, and its value is $\det A / \det A[S_r, S_r]$. The determinant factors cancel when probability is multiplied by final trace.

In LU, the product of the pivot values equals the determinant of the selected $r \times r$ block in the ordered row and column lists. Its squared modulus is the numerator of the history probability. The only remaining residual entry has squared modulus $|\det A|^2/|\det A[I_r, J_r]|^2$; permutations change only a sign before taking the modulus. Cancellation gives (20). Summing history contributions proves both identities. No independence of the LU row and column choices is being assumed. □

6 Counting the surviving histories

First identify the optimal error scale. The determinant and inverse of the family satisfy

$$\det A = \epsilon^{r(r+1)/2}, \qquad \epsilon^r A^{-1} = L^{-\mathsf{T}} \operatorname{diag}(\epsilon^r, \epsilon^{r-1}, \ldots, 1) L^{-1} \longrightarrow e_n e_n^{\mathsf{T}}. \tag{21}$$

The largest eigenvalue on the right-hand side tends to 1, whence

$$\frac{\lambda_n(A)}{\epsilon^r} \longrightarrow 1. \tag{22}$$

Since A has order $r + 1$ and is positive definite, $\tau_r(A) = \lambda_n(A)$ and $\left\| A - A_r^{\mathrm{best}} \right\|_{\mathrm{F}}^2 = \lambda_n(A)^2$. Using (21) in the first history identity gives

$$\frac{\mathbb{E} \operatorname{tr} R_r}{\lambda_n(A)} = \frac{\epsilon^r}{\lambda_n(A)} \sum_{(j_1,\ldots,j_r)} \prod_{s=0}^{r-1} \frac{\epsilon^s}{T_s(S_s)}. \tag{23}$$

By Lemma 7, the product for a fixed history tends to 1 precisely when

$$s + 1 \notin S_s \qquad (0 \le s \le r - 1), \tag{24}$$

and otherwise tends to 0. The condition at $s = 0$ is vacuous.

Condition (24) is equivalent to

$$j_u \in \{1, \dots, u, n\} \setminus \{j_1, \dots, j_{u-1}\} \qquad (1 \le u \le r). \tag{25}$$

Indeed, choosing a label $q \in \{2, \dots, r\}$ before position q violates (24) at $s = q - 1$. Conversely, if every such label is chosen no earlier than its own position, no prefix can contain its next label. Label $n = r + 1$ has no corresponding normalizer constraint, because the normalizers stop at $s = r - 1$.

At position u, the allowed set before removing earlier choices has $u + 1$ elements. All $u - 1$ earlier choices lie in it. Thus there are exactly two choices at every position, and exactly 2^r surviving histories. The sum in (23) is finite for each fixed r. Together with (22), this proves the Cholesky limit.

For LU, the corresponding normalized identity is

$$\frac{\mathbb{E}\,\|B_r\|_{\mathrm{F}}^2}{\lambda_n(A)^2} = \frac{\epsilon^{2r}}{\lambda_n(A)^2} \sum_{\substack{(i_1,\dots,i_r)\\(j_1,\dots,j_r)}} \prod_{s=0}^{r-1} \frac{\epsilon^{2s}}{Q_s(I_s, J_s)}. \tag{26}$$

Lemma 7 says that a summand tends to 1 exactly when both the row history and the column history satisfy (24); it otherwise tends to 0. Each list has 2^r possibilities, giving 4^r ordered pairs. This is a factorization of the limiting indicator and a combinatorial count, not a claim that the algorithm samples its row and column histories independently. Equation (22) now proves the LU limit and completes the proof of Theorem 1. $\square$

7 Unit-diagonal inputs by row replication

The examples of order $r + 1$ are not unit-diagonal. Nevertheless, requiring a unit diagonal does not improve either worst-case constant when arbitrary matrix dimensions are allowed.

Proposition 9 (Correlation-matrix sharpness)**.** *For each fixed $r \ge 1$, both suprema in Corollary 2 remain unchanged when inputs are restricted to real, entrywise-positive, positive-definite matrices with every diagonal entry equal to* 1*.*

Proof. First let A be any rational positive-definite matrix of order n with positive entries. Choose $c > 0$ rational so that $m_i = A_{ii}/c$ is a positive integer for every i. Put $N = \sum_i m_i$. Partition the N row labels into groups of sizes $m_1, \dots, m_n$, and construct $P \in \mathbb{R}^{N \times n}$ by assigning each row in group i the vector $e_i^{\mathsf{T}}/\sqrt{m_i}$. Then $P^{\mathsf{T}}P = I_n$, and

$$C = c^{-1} P A P^{\mathsf{T}}$$

is positive semidefinite, entrywise positive, and unit-diagonal. Its nonzero eigenvalues are those of A/c.

The two pivoting processes on C exactly reproduce those on A after aggregating pivot labels by group. More precisely, if the current residual has the form $c^{-1} P B P^{\mathsf{T}}$, a Cholesky pivot in group i has aggregate probability

$$\frac{m_i B_{ii}/(c m_i)}{\operatorname{tr} B/c} = \frac{B_{ii}}{\operatorname{tr} B}.$$

For LU, the aggregate probability of a row in group i and a column in group j is

$$\frac{m_i m_j |B_{ij}|^2/(c^2 m_i m_j)}{\|B\|_{\mathrm{F}}^2/c^2} = \frac{|B_{ij}|^2}{\|B\|_{\mathrm{F}}^2}.$$

Substituting either pivot into its rank-one update shows that the new residual is $c^{-1}PB_{\mathrm{new}}P^{\mathsf{T}}$. These assertions also respect zero pivot probabilities. Thus the trace error is scaled by $1/c$ and the squared Frobenius error by $1/c^2$, exactly as are the respective optimal rank-r errors. Both expected-error ratios are unchanged.

To obtain positive definiteness, put

$$C_\delta = \frac{C + \delta I_N}{1+\delta}, \qquad \delta > 0.$$

This matrix is positive definite, entrywise positive, and unit-diagonal. Fix a finite stopping rank r for which the optimal error of C is positive. Every pivot history having positive probability for C has nonzero pivots at all of its steps. Along each such history, the pivot values, probabilities, and terminal residuals for C_δ converge to those for C, because the rank-one updates are rational functions whose denominators remain nonzero. There are only finitely many histories. Summing these convergent contributions and discarding all other, nonnegative contributions proves

$$\liminf_{\delta \downarrow 0} F_r(C_\delta) \geq F_r(C),$$

where F_r is either expected-error ratio. Here the optimal-error denominators converge to their positive values for C, by continuity of the eigenvalues. Full continuity of the LU expectation at zero pivot entries is not needed.

Finally choose rational t in Theorem 1 to approach the desired sharp constant, apply the replication construction to $A_r(t)$, and then take δ sufficiently small. The known universal upper bounds complete the proof. The replicated dimension N is not bounded uniformly in t. □

8 Exact finite certificates and scope

The limiting proof establishes a counterexample for every proposed smaller uniform factor. The accompanying code also supplies explicit finite rational certificates. These are useful because the matrices are deliberately ill-conditioned, making an ordinary double-precision eigensolve unreliable.

Let $v = L^{-\mathsf{T}} e_n$ and $\nu = \|v\|_2^2$. From (21),

$$\epsilon^r A^{-1} \succeq vv^{\mathsf{T}}, \qquad \lambda_n(A) \leq \frac{\epsilon^r}{\nu}.$$

For $0 \leq s \leq r-1$, let $\mathcal{G}_s$ consist of all s-element subsets of $\{1, \ldots, s, n\}$, and define the exact rational maxima

$$m_s = \max_{S \in \mathcal{G}_s} \frac{T_s(S)}{\epsilon^s}, \qquad q_s = \max_{I,J \in \mathcal{G}_s} \frac{Q_s(I,J)}{\epsilon^{2s}}.$$

Every history counted in (25) has its sth prefix in $\mathcal{G}_s$. Retaining only these histories in (23) and (26) gives

$$\frac{\mathbb{E}\,\mathrm{tr}\,R_r}{\lambda_n(A)} \geq \frac{2^r \nu}{\prod_{s=0}^{r-1} m_s}, \qquad \frac{\mathbb{E}\,\|B_r\|_{\mathrm{F}}^2}{\lambda_n(A)^2} \geq \frac{4^r \nu^2}{\prod_{s=0}^{r-1} q_s}. \tag{27}$$

For a finite certificate it suffices to verify that these retained pivot histories are valid; nonsingularity of their prefix and terminal blocks is checked exactly. Omitted histories contribute nonnegative error.

The file `exact_certificate.py` evaluates (27) with Python `Fraction` arithmetic. The separate file `exact_enumeration.py` recursively executes all actual rank-one pivot updates for small dimensions, without using the history cancellation formula to compute expectations. It encloses the optimal error using

$$\frac{\epsilon^r}{\operatorname{tr}(\epsilon^r A^{-1})} \le \lambda_n(A) \le \frac{\epsilon^r}{\nu}.$$

Both endpoints and both error expectations are rational, so the resulting ratio enclosures are rigorous. Selected lower bounds at $t = 1/100$ are shown below; the displayed decimals are rounded downward, and the exact fractions are stored in the accompanying JSON records.

r	Certificate	Cholesky ratio >	2^r	LU ratio >	4^r
2	All pivot updates	3.99995644	4	15.99965154	16
3	All pivot updates	7.99957072	8	63.99313178	64
8	Retained histories	255.85420697	256	65461.37522779	65536

The rank-eight certificate uses $n = 9$ and $\epsilon = (1/100)^{81}$. The unusually small scales are part of an exact-arithmetic worst-case construction, not a proposed practical test in double precision.

The small-dimensional family has unrestricted diagonal entries; Proposition 9 separately covers the unit-diagonal setting in unrestricted dimension. No bounded-condition-number claim is made. These results do not resolve how many pivots beyond r suffice for a $(1 + \delta)$-relative trace approximation, the separate oversampling question RA-01. In particular, after $r + 1$ pivots the matrices in this note are recovered exactly. Zero-padding embeds the lower-bound family in any larger dimension without altering its nonzero-pivot process, but does not turn this same-rank result into an oversampling lower bound.

References

[1] Y. Chen, E. N. Epperly, J. A. Tropp, and R. J. Webber. *Randomly pivoted Cholesky: Practical approximation of a kernel matrix with few entry evaluations.* Communications on Pure and Applied Mathematics, 78(5):995–1041, 2025. doi:10.1002/cpa.22234. The universal same-rank bound is Lemma 5.5.

[2] M. A. Gilles and H. Wilber. *Low-Rank Approximation by Randomly Pivoted LU.* arXiv:2601.22344, 2026. `https://arxiv.org/abs/2601.22344`. The equation numbering here follows the author-posted full text dated 2 February 2026. The sampling rule is equation (3), the bound is Theorem 3, and the proposed 2^r improvement appears immediately after it.

[3] *Open Problems in Numerical Linear Algebra*, randomized and low-rank approximation category, entries RA-02 and RA-03. `https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/randomized-and-low-rank-approximation`. The repository-audit record accompanying this note distinguishes the retrieved category listing from source-file and pull-request checks that could not be completed. The mathematical statements proved here are specified in full above and do not depend on a repository status label.